$$\begin{aligned}
\langle \dot{\sigma_x} \rangle &= \frac{-i\omega}{2} \langle [\sigma_x, \sigma_z] \rangle \\
\langle \dot{\sigma_y} \rangle &= \frac{-i\omega}{2} \langle [\sigma_y, \sigma_z] \rangle \\
\langle \dot{\sigma_z} \rangle &= \frac{-i\omega}{2} \langle [\sigma_z, \sigma_z] \rangle . \qquad (4.25)
\end{aligned}$$

The things we are computing on the left side of the equations are supposed to be real quantities. The factor *i* in these equations seems like trouble. Fortunately, the commutation relations between σ_x, σ_y, and σ_z will save the day. By plugging in the Pauli matrices from Eq. 3.20, it's easy to verify that

$$\begin{aligned}
[\sigma_x, \sigma_y] &= 2i\sigma_z \\
[\sigma_y, \sigma_z] &= 2i\sigma_x \\
[\sigma_z, \sigma_x] &= 2i\sigma_y . \qquad (4.26)
\end{aligned}$$

Each of these equations also has an *i*, which will cancel the *i* in Eqs. 4.25. Notice that the factors of 2 also cancel, resulting in some very simple equations: